$$A \coloneqq \text{Taking 2 Red}$$
$$B \coloneqq \text{Taking 2 Black}$$
$$C \coloneqq \text{Taking 1 of each}$$

$$R \coloneqq \text{Number of Red balls}$$
$$B \coloneqq \text{Number of Black balls}$$

$$P(a,b) \coloneqq P(R=a, B=b)$$

$$P(A) = \begin{cases} \frac{\binom{R}{2}}{\binom{R+B}{2}} \text{ if } R \geq 2 \\ 0 \text{ otherwise} \end{cases}$$

$$P(B) = \begin{cases} \frac{\binom{B}{2}}{\binom{R+B}{2}} \text{ if } B \geq 2 \\ 0 \text{ otherwise} \end{cases}$$

$$P(C) = \begin{cases} \frac{\binom{R}{1} \cdot \binom{B}{1}}{\binom{R+B}{2}} \text{ if } R \geq 1 \land B \geq 1 \\ 0 \text{ otherwise} \end{cases}$$

$$P(a,b) = \begin{cases} 1 \text{ if } a == 0 \\ 0 \text{ if } b == 0 \\ f(a,b) \text{ otherwise} \end{cases}$$

$$\begin{aligned} f(a,b) = P(A;a,b) \cdot P(a-2,b) + \\ +P(B;a,b) \cdot P(a,b) \\ +P(C;a,b) \cdot P(a,b-1) \end{aligned}$$

$$\binom{n}{1} = n$$

$$\begin{aligned}P(2,2) = P(A;2,2)\cdot P(0,2) + \\ +P(B;2,2)\cdot P(2,2) \\ +P(C;2,2)\cdot P(2,1)\end{aligned}$$

$$P(0,2) = 1$$

$$\begin{aligned}P(2,1) = P(A;2,1)\cdot P(0,1) + \\ +P(B;2,1)\cdot P(2,1) \\ +P(C;2,1)\cdot P(2,0)\end{aligned}$$

$$\begin{aligned}P(0,1) &= 1 \\ P(2,0) &= 0 \\ P(B;2,1) &= 0\end{aligned}$$

$$\begin{aligned}\therefore P(2,1) &= P(A;2,1)\cdot 1 + \\ &+0\cdot P(2,1) \\ &+P(C;2,1)\cdot 0 \\ P(2,1) &= P(A;2,1) \\ &= \frac{\binom{2}{2}}{\binom{3}{2}} \\ &= \frac{1}{3}\end{aligned}$$

$$\begin{aligned}P(B;2,2) &= \frac{\binom{2}{2}}{\binom{4}{2}} \\ &= \frac{1}{6} \\ P(A;2,2) &= \frac{\binom{2}{2}}{\binom{4}{2}} \\ &= \frac{1}{6} \\ P(C;2,2) &= \frac{2\cdot 2}{\binom{4}{2}} \\ &= \frac{4}{6}\end{aligned}$$

$$\begin{aligned}\therefore P(2,2) &= \frac{1}{6}\cdot 1 + \\ &+\frac{1}{6}\cdot P(2,2) \\ &+\frac{4}{6}\cdot\frac{1}{3} \\ \equiv P(2,2) &= \frac{7}{15} = 0.4(6)\end{aligned}$$

$\blacksquare$

$$
\begin{aligned}
f(a,b) = P(A;a,b)\cdot f(a-2,b) &+ \\
+P(B;a,b)\cdot f(a,b)& \\
+P(C;a,b)\cdot f(a,b-1)&
\end{aligned}
$$

$$
\begin{aligned}
\alpha &= P(A;a,b) \\
\beta &= P(B;a,b) \\
\gamma &= P(C;a,b) \\
y &= f(a-2,b) \\
x &= f(a,b) \\
z &= f(a,b-1)
\end{aligned}
$$

$$
\begin{aligned}
x &= \alpha\cdot y + \beta\cdot x + \gamma\cdot z \\
\gamma z &= (1-\beta)x - \alpha y \\
x &= \frac{\gamma z + \alpha y}{1-\beta}
\end{aligned}
$$